Final Project

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2025-03-11

Introduction

Nutrition is an essential to everyday life, but the enjoyment of food and beverages is equally important. Understanding the factors the shape food and drink enjoyment is essential for both consumer behavior and product development. The ingredients receive much attention, but the way the product is experienced might be equally as important. Companies might need to redesign their product not by ingredients, but by design. Examining variables like pouring and consumption might influence sensory perception giving insights into how presentation affects taste and overall enjoyment.

We all claim to have a favorite food or drink, but the way we consume them could play a greater role than the ingredients themselves. Taste is considered the primary determinant of preference, however sensory perception is influenced by factors beyond flavor. Texture, temperature, smell, audio cues, and the environment all can impact the quality of consumables. The environment and sounds we hear prior might influence our sensory expectations. The audio of pouring liquid and fizz of a carbonated drink might make a drink "taste" better by alternating the perceived freshness, sweetness, and overall enjoyment (Spence, et. all).

In carbonated beverages, the bubble and foam relationship with chemical components influences sensory characteristics such as aroma release, mouthfeel, and perception of tastes and aromas (Gonzalez, et. all). Research suggests consumers evaluate carbonated beverages more on the levels of carbonation, foam stability, and bubble activity more than the taste. This is why a flat carbonated beverage is perceived as low quality (Gonzalez, et. all). Bubbles are important because they are associated with how much carbonation is in the beverage. A previous study concluded that ratings of carbonation bite under two pressures (one that caused bubbles and one that didn't) were the same. The bubbles were not required for carbonation bite, but modulated perceived bite (Wise, et. all). This is why the amount of carbonation (and perceived carbonation) is important. The way carbonation is retained or lost through pouring and drinking methods could significantly influence consumer enjoyment.

Methods

In this study I am using a factorial design with blocking to examine whether the method of pouring and method of consumption influences the perceived taste of Coca-Cola. I have a total of 7 participants that each will consume Coca-Cola in six different ways. The participants recruited were my friends who were not available at the same time, so the data was collected over different days. The factors of interest are: the method of pouring and method of consumption. The participants are each a block because I am not interested in how the participants score compared to each other. The three methods of pouring are: Down the side of the glass, straight onto the center of a glass, or left in the can. The two methods of consumption are using a straw and straight from the container.

When conducting the experiment, I used random number generation to randomly assign the order of the methods for each participant. Before every drink, I had the participants drink cold water to "wash" their mouth. I had the participants wait five minutes between drinks. I asked each participant to rate their drink

by taste on a scale from 1 - 10 (1 = worst; 10 = best). I put their responses in an excel sheet and manually put those results into R Studio.

Since the experiment was over multiple days, there was potential issues that longer aged soda might taste different. So each can was properly stored in a refrigerator and not touched until it was ready to be used. Each participant had their drinks come from one can. Since the soda was refrigerated and the participants waited five minutes between drinks, there could be a potential bias for the tests that were conducted later for each experiment. The soda slowly became warmer as time went on and the open can might influence the amount of carbonation. The order they drank was random to hopefully mitigate these issues.

My null hypothesis (H_0) is that none of my interested factors (pouring method, consumption method, or combination) contribute to score. My alternative hypothesis (H_A) is that at least one interested factor affects score. I am using $\alpha = 0.05$ for my significance level. Later I will be using three separate hypothesis tests so I adjusted for the multiple comparisons by using Bonferroni correction. To do this I took my α level and divided by the number of tests. So the calculation was 0.05 / 3 = 0.01667. Now if any p-value is less than 0.01667, then it is statistically significant.

The first test to run is a sample size calculation to see what an appropriate sample size should be. Unfortunately my data doesn't have a large enough sample size so if this experiment was replicated it should have a larger sample size.

Then I visualized the data by using a jitter plot and interaction plot. The jitter plot is a good starting point to see if I can depict any trends visually. I might be able to make some inferences on the data before I actually run any tests. The interaction plot gives me the averages for every combination of factors of interest. For the plot I can visually see if it appears that the interaction between the factors is significant.

Afterwards, I ran a linear model on the data to get a F-statistic and P-value to see if any combination of factors contributes to the model. To actually know what factors are important, I used an anova test on two linear models for each factor of interest. The first model was always the full model and the second model had all factors of interests minus the factor I was interested in. First I tested if the pour method was significant, then if the consumption method was significant, and finally I tested if the interaction between them was significant.

After testing all factors I need to know if my results are correct or biased. To do this I need to look at the residuals. The residuals should appear to be normally distributed without an unusual amount of outliars. If the residuals are not valid, then permutation tests (randomly shuffling the data) are required for each factor to properly know if there is statistically enough evidence. Unfortunately, the assumptions for my data was not valid so after using linear models and anova, I use permutation tests.

Results

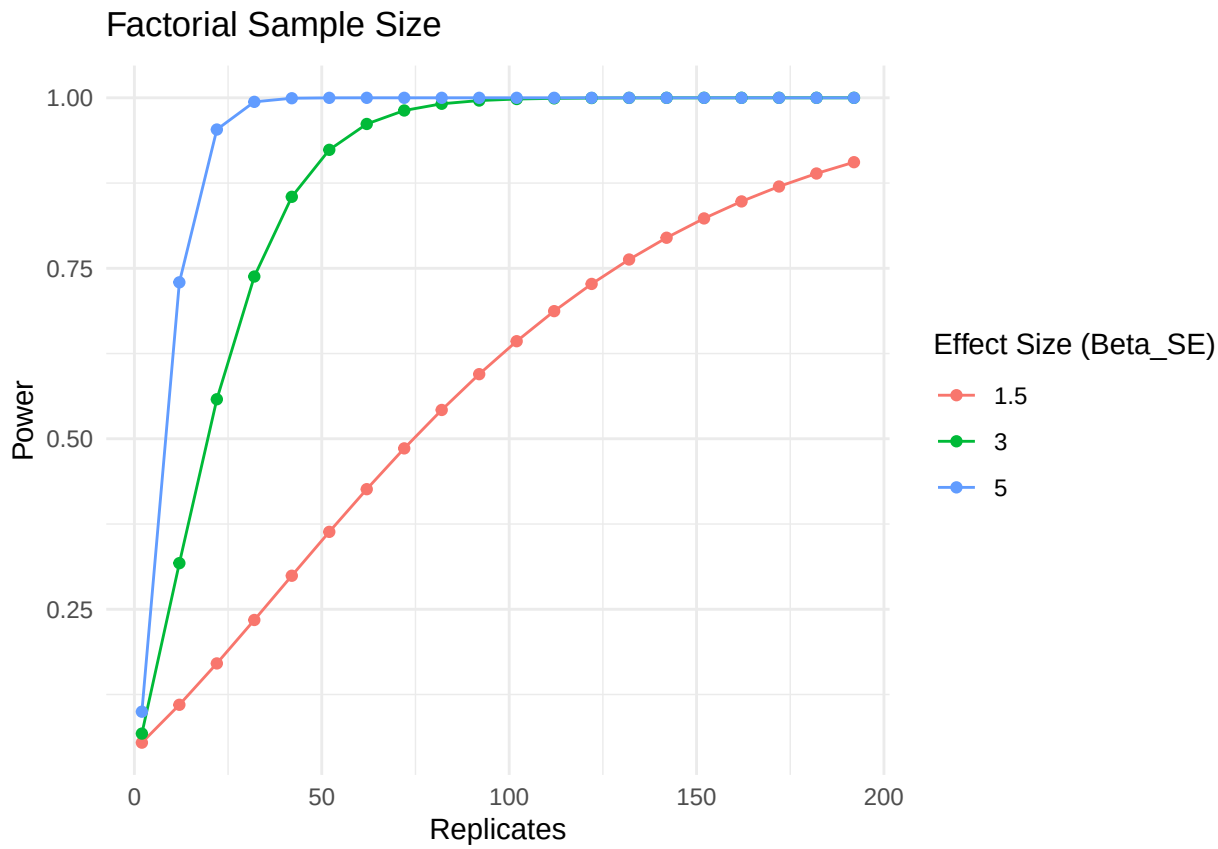
I took my excel file and manually put the values into R Markdown.

```r
set.seed(275690)

P1 <- c(7, 6, 5, 6, 8, 7)
P2 <- c(6, 5, 4, 5, 7, 6)
P3 <- c(8, 7, 6, 7, 10, 8)
P4 <- c(5, 6, 4, 5, 6, 5)
P5 <- c(7, 6, 5, 6, 8, 7)
P6 <- c(6, 5, 4, 5, 7, 6)
P7 <- c(8, 7, 6, 7, 9, 8)

soda_df <- data.frame(
  score = c(P1, P2, P3, P4, P5, P6, P7),
```

```r
  pour = c(rep(c("Down Side", "Center", "Can"), 7, each = 2)),
  consumption = c(rep(c("Straw", "Direct"), 21)),
  participant = c(rep("Participant 1", length(P1)), rep("Participant 2", length(P2)),
            rep("Participant 3", length(P3)), rep("Participant 4", length(P4)),
            rep("Participant 5", length(P5)), rep("Participant 6", length(P6)),
            rep("Participant 7", length(P7)))
)

library(ggplot2)
suppressMessages(library(dplyr))
```

First, I created a sample size graph to see what an appropriate sample size should be for my data.

```r
suppressWarnings(library(pwr))

replicates <- seq(2, 200, by=10)
beta_se_values <- c(1.5, 3, 5)

results <- data.frame()
for (beta_se in beta_se_values) {
  for (n in replicates) {
    power_result <- pwr.anova.test(k = 3, n = n, f = beta_se / 10, sig.level = 0.05)$power

    results <- rbind(results, data.frame(
      replicates = n,
      power = power_result,
      beta_se = factor(beta_se)
    ))
  }
}

ggplot(results, aes(x = replicates, y = power, color = beta_se, group = beta_se)) +
  geom_line() +
  geom_point() +
  theme_minimal() +
  labs(title = expression("Factorial Sample Size"),
       x = "Replicates",
       y = "Power",
       color = "Effect Size (Beta_SE)") +
  theme(legend.position = "right")
```



Unfortunately in my data a large sample size is required to high a strong enough power to avoid Type I errors. If my Beta Standard Error was around 3, I would need a sample size of approximately 50 to have a power of 0.80. If this experiment was replicated, a larger sample size would be required to make the tests more acceptable.

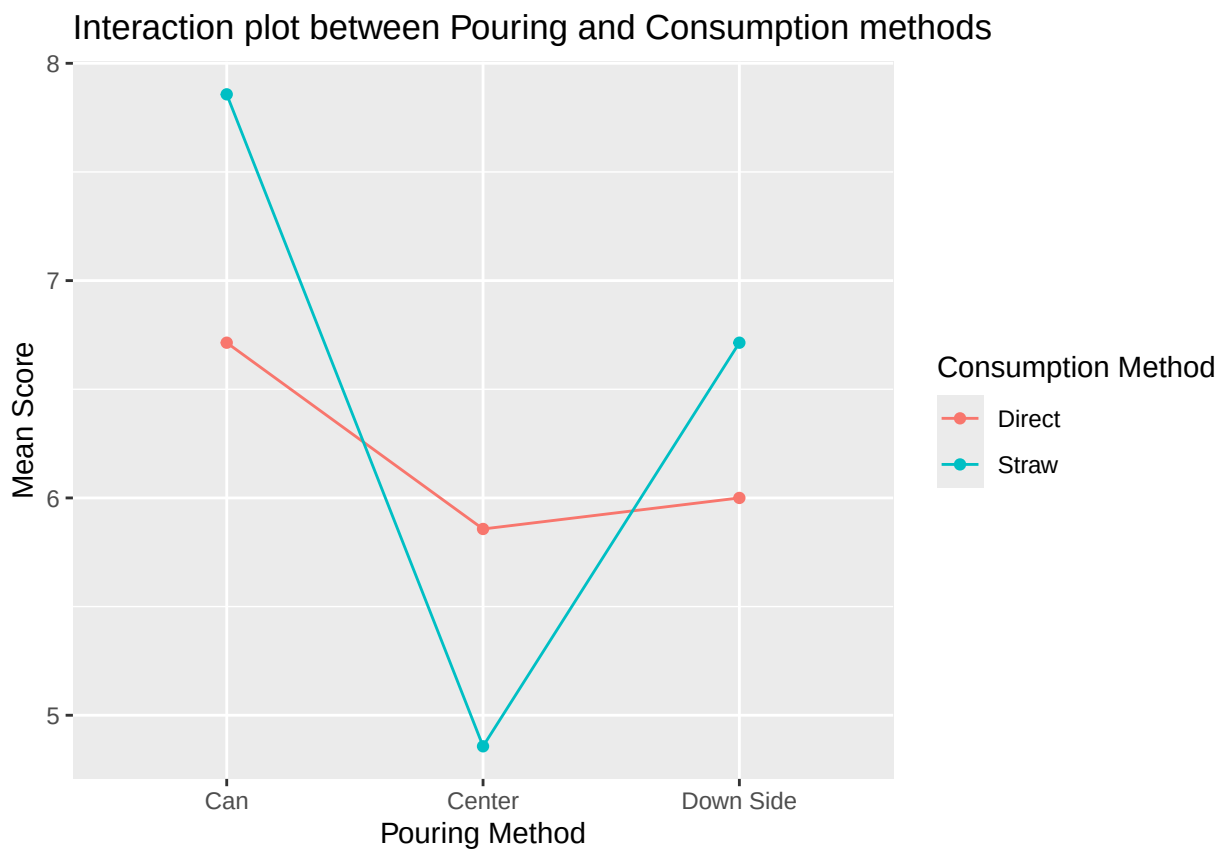
To visualize the data, I create a jitter plot to see if I can depict any trends.

```
ggplot(data = soda_df, mapping = aes(x = pour, y = score, color = consumption)) +
  geom_jitter(width = 0.15, height = 0) +
  labs(
    x = "Pouring Method",
    color = "Consumption Method",
    title = "Distribution of Scores by Pouring and Consumption method"
  )
```

After visualizing the jitter plot, it appears drinking from the can results in the highest score. Pouring the soda in the center of the glass resulted in the lowest overall score with the highest value being seven. The only two scores to get above a score of eight was when drinking from a can + straw. Scores after the soda was poured down the side were slightly above center but way below drinking straight from the can.

To see if there is an interaction between the consumption and pouring methods, I created a interaction plot by looking at the means of each.

```
soda_summary <- soda_df %>%
  group_by(pour, consumption) %>%
  summarise(mean_score = mean(score), .groups = "drop")

ggplot(soda_summary, aes(x = pour, y = mean_score, color = consumption, group = consumption)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Pouring Method",
    y = "Mean Score",
    color = "Consumption Method",
    title = "Interaction plot between Pouring and Consumption methods"
  )
```



After visualizing the interaction plot, drinking from a straw usually results in a better score except when the soda was poured in the middle. Drinking directly is mostly linear, with all the means being within one score of each other. Drinking from a straw has a wider variance with drinking from the center having a score below five. Drinking with a straw out of the can resulted in the highest mean score around eight.

To actually compute the data, running a linear model will be the best to test if any of the factors have an affect on the model

```
library(knitr)
soda_model <- lm(score ~ pour + consumption + pour * consumption + participant, data = soda_df)
soda_model_summary <- summary(soda_model)

df_residual <- soda_model_summary$df[2]  # Residual degrees of freedom
r_squared <- soda_model_summary$r.squared  # R-squared
adj_r_squared <- soda_model_summary$adj.r.squared  # Adjusted R-squared
f_value <- soda_model_summary$fstatistic[1]  # F-statistic
p_value <- pf(f_value, soda_model_summary$fstatistic[2], soda_model_summary$fstatistic[3], lower.tail =

model_summary_table <- data.frame(
  Metric = c("Residual DF", "R-squared", "Adjusted R-squared", "F-statistic", "P-value"),
  Value = round(c(df_residual, r_squared, adj_r_squared, f_value, p_value), 4)
)

# Display table
kable(model_summary_table, digits = 4, caption = "Soda Linear Model Summary")
```

Table 1: Soda Linear Model Summary

Metric	Value
Residual DF	30.0000
R-squared	0.9501
Adjusted R-squared	0.9318
F-statistic	51.8872
P-value	0.0000

After running a linear model, I get a P-value of approximately zero and conclude at $\alpha = 0.01667$ that at least one factor or combination does affect the participants score.

First I want to check if the block (participant) is the only factor affecting score. Since participants are my blocks, I hope this is not the case. To check this I ran anova on two linear models: the full model and a model that individual checks score and participants.

```
soda_participant <- lm(score ~ participant, data = soda_df)

anova1 <- anova(soda_participant, soda_model)

df_residual_1 <- anova1$Df[1]  # Residual DF for Model 1
df_residual_2 <- anova1$Df[2]  # Residual DF for Model 2
rss_1 <- anova1$RSS[1]  # RSS for Model 1
rss_2 <- anova1$RSS[2]  # RSS for Model 2
sum_sq <- anova1$"Sum of Sq"[2]  # Sum of Squares for the added predictors
f_value <- anova1$F[2]  # F-statistic
p_value <- anova1$"Pr(>F)"[2]  # P-value

anova_summary_table <- data.frame(
  Metric = c("Residual DF (Model 1)", "Residual DF (Model 2)",
             "Residual Sum of Squares (Model 1)", "Residual Sum of Squares (Model 2)",
             "Sum of Squares for Added Predictors", "F-statistic", "P-value"),
  Value = round(c(df_residual_1, df_residual_2, rss_1, rss_2, sum_sq, f_value, p_value), 4)
)

# Display the table
kable(anova_summary_table, digits = 4, caption = "ANOVA Model Comparison of participant blocks")
```

Table 2: ANOVA Model Comparison of participant blocks

Metric	Value
Residual DF (Model 1)	NA
Residual DF (Model 2)	5.0000
Residual Sum of Squares (Model 1)	39.6667
Residual Sum of Squares (Model 2)	3.7619
Sum of Squares for Added Predictors	35.9048
F-statistic	57.2658
P-value	0.0000

After running anova on the two linear models, I get a p-value of approximately zero and conclude that some combination of other factors (pour, consumption, and their interaction) also have an affect on score. This works with the experiment because the participants should not be the only factor in the model.

To check if the pouring method and its interaction affects score, I ran an anova on two linear models: The full model and one that only takes the consumption method and participants into account.

```
soda_no_pour <- lm(score ~ consumption + participant, data = soda_df)

anova2 <- anova(soda_no_pour, soda_model)

df_residual_1 <- anova2$Df[1]  # Residual DF for Model 1
df_residual_2 <- anova2$Df[2]  # Residual DF for Model 2
rss_1 <- anova2$RSS[1]  # RSS for Model 1
rss_2 <- anova2$RSS[2]  # RSS for Model 2
sum_sq <- anova2$"Sum of Sq"[2]  # Sum of Squares for the added predictors
f_value <- anova2$F[2]  # F-statistic
p_value <- anova2$"Pr(>F)"[2]  # P-value

anova_summary_table <- data.frame(
  Metric = c("Residual DF (Model 1)", "Residual DF (Model 2)",
             "Residual Sum of Squares (Model 1)", "Residual Sum of Squares (Model 2)",
             "Sum of Squares for Added Predictors", "F-statistic", "P-value"),
  Value = round(c(df_residual_1, df_residual_2, rss_1, rss_2, sum_sq, f_value, p_value), 4)
)

# Display the table
kable(anova_summary_table, digits = 4, caption = "ANOVA Model Comparison of pour method")
```

Table 3: ANOVA Model Comparison of pour method

Metric	Value
Residual DF (Model 1)	NA
Residual DF (Model 2)	4.0000
Residual Sum of Squares (Model 1)	38.8095
Residual Sum of Squares (Model 2)	3.7619
Sum of Squares for Added Predictors	35.0476
F-statistic	69.8734
P-value	0.0000

After running the model I get a P-value of approximately zero and conclude that there is statistically significant evidence that the pouring method and its interaction does have an affect on score.

Just like testing the pour method, to test if the consumption method has an affect on score, I ran an anova test on the two linear models: the full model and the model that only includes the pour method and participants.

```
soda_no_consumption <- lm(score ~ pour + participant, data = soda_df)

anova3 <- anova(soda_no_consumption, soda_model)

df_residual_1 <- anova3$Df[1]  # Residual DF for Model 1
df_residual_2 <- anova3$Df[2]  # Residual DF for Model 2
rss_1 <- anova3$RSS[1]  # RSS for Model 1
rss_2 <- anova3$RSS[2]  # RSS for Model 2
sum_sq <- anova3$"Sum of Sq"[2]  # Sum of Squares for the added predictors
```

```r
f_value <- anova3$F[2]  # F-statistic
p_value <- anova3$"Pr(>F)"[2]  # P-value

anova_summary_table <- data.frame(
  Metric = c("Residual DF (Model 1)", "Residual DF (Model 2)",
             "Residual Sum of Squares (Model 1)", "Residual Sum of Squares (Model 2)",
             "Sum of Squares for Added Predictors", "F-statistic", "P-value"),
  Value = round(c(df_residual_1, df_residual_2, rss_1, rss_2, sum_sq, f_value, p_value), 4)
)

# Display the table
kable(anova_summary_table, digits = 4, caption = "ANOVA Model Comparison of consumption method")
```

Table 4: ANOVA Model Comparison of consumption method

Metric	Value
Residual DF (Model 1)	NA
Residual DF (Model 2)	3.0000
Residual Sum of Squares (Model 1)	13.6190
Residual Sum of Squares (Model 2)	3.7619
Sum of Squares for Added Predictors	9.8571
F-statistic	26.2025
P-value	0.0000

After running the test, I get a P-value of approximately zero and conclude that there is statistically significant evidence that the consumption method and its interaction does have an affect on score.

To test if the interaction between the pour and consumption method is significant I run another anova on two linear models: the full model and a model including everything but the interaction.

```r
soda_test_int <- lm(score ~ pour + consumption + participant, data = soda_df)

anova4 <- anova(soda_test_int, soda_model)

df_residual_1 <- anova4$Df[1]  # Residual DF for Model 1
df_residual_2 <- anova4$Df[2]  # Residual DF for Model 2
rss_1 <- anova4$RSS[1]  # RSS for Model 1
rss_2 <- anova4$RSS[2]  # RSS for Model 2
sum_sq <- anova4$"Sum of Sq"[2]  # Sum of Squares for the added predictors
f_value <- anova4$F[2]  # F-statistic
p_value <- anova4$"Pr(>F)"[2]  # P-value

anova_summary_table <- data.frame(
  Metric = c("Residual DF (Model 1)", "Residual DF (Model 2)",
             "Residual Sum of Squares (Model 1)", "Residual Sum of Squares (Model 2)",
             "Sum of Squares for Added Predictors", "F-statistic", "P-value"),
  Value = round(c(df_residual_1, df_residual_2, rss_1, rss_2, sum_sq, f_value, p_value), 4)
)

# Display the table
kable(anova_summary_table, digits = 4, caption = "ANOVA Model Comparison of interaction")
```

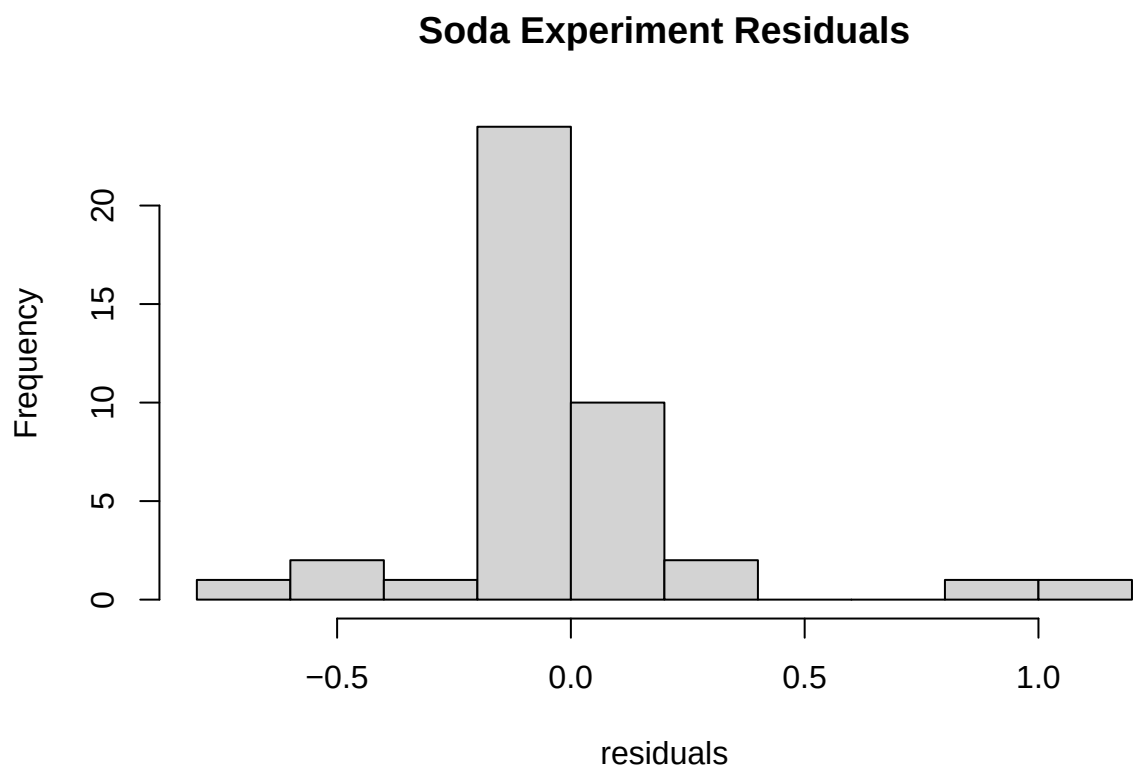
Table 5: ANOVA Model Comparison of interaction

Metric	Value
Residual DF (Model 1)	NA
Residual DF (Model 2)	2.0000
Residual Sum of Squares (Model 1)	12.7619
Residual Sum of Squares (Model 2)	3.7619
Sum of Squares for Added Predictors	9.0000
F-statistic	35.8861
P-value	0.0000

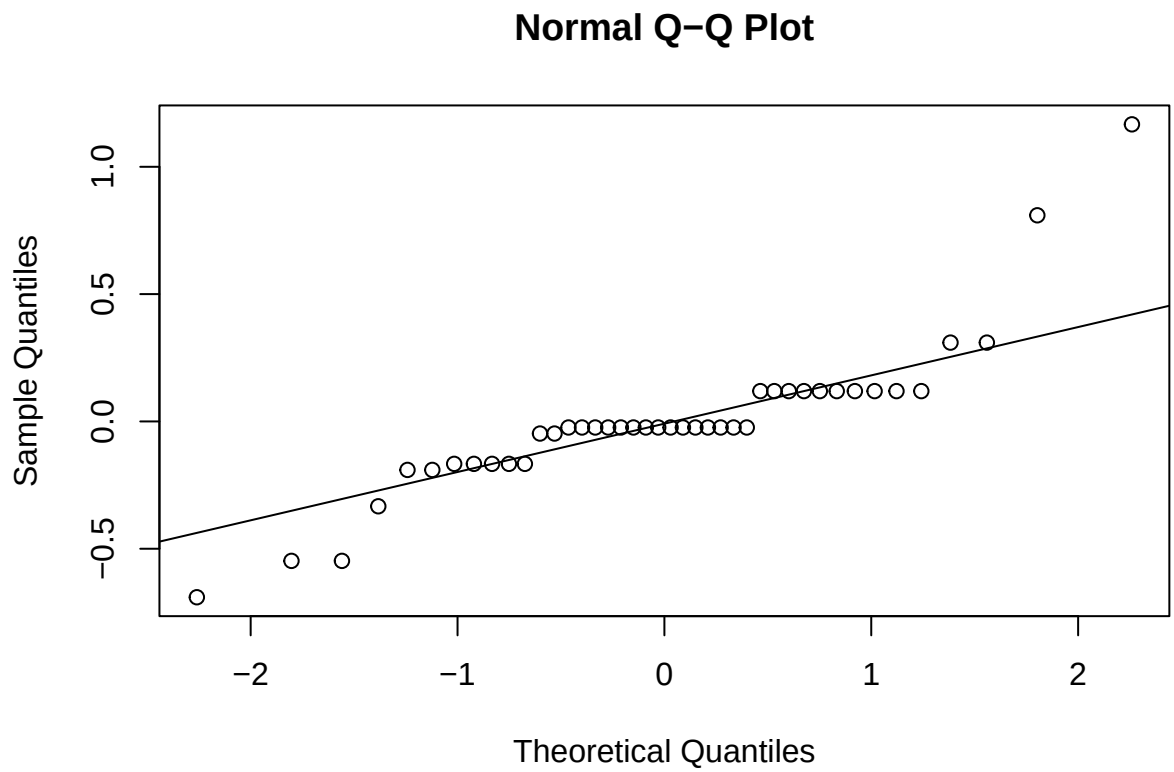
After running the model I get a P-value of approximently zero and conclude that there is statistically significant evidence that the interaction between the pouring and consumption method does have an affect on score.

Now I have to check whether the model is a good fit for the data. To check this I will create a histogram of the residuals to check for normality, a QQ-plot to check if the residuals are adequate, and run a Shapiro-Wilk Test.

```
hist(soda_model$residuals, xlab = "residuals", main = "Soda Experiment Residuals")
```




```
qqnorm(soda_model$residuals)
qqline(soda_model$residuals)
```

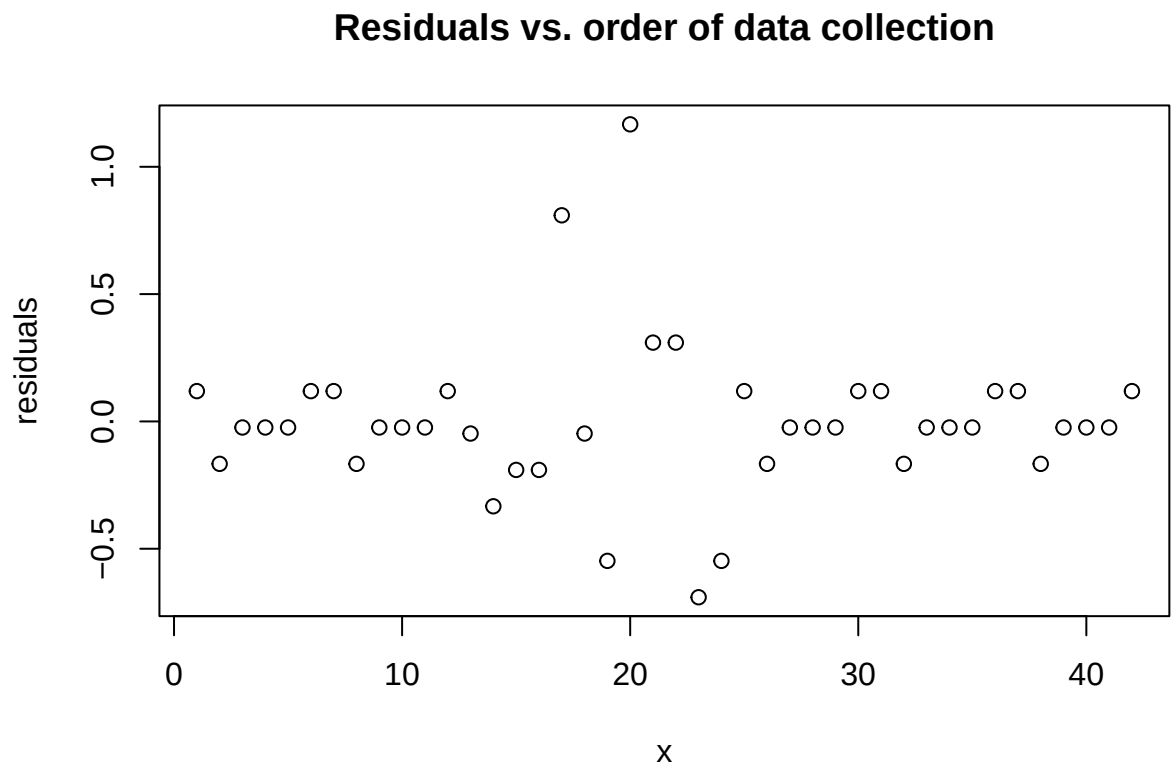



```r
shapiro_test<- shapiro.test(soda_model$residuals)

w_statistic <- shapiro_test$statistic
p_value <- shapiro_test$p.value

shapiro_summary_table <- data.frame(
  Metric = c("Shapiro-Wilk W Statistic", "P-value"),
  Value = round(c(w_statistic, p_value), 6)
)

# Display the table
kable(shapiro_summary_table, digits = 6, caption = "Shapiro-Wilk Normality Test Summary")
```

Table 6: Shapiro-Wilk Normality Test Summary

	Metric	Value
W	Shapiro-Wilk W Statistic	0.811034
	P-value	0.000008

```r
x <- 1:length(soda_model$residuals)
plot(soda_model$residuals ~ x, ylab="residuals",
main="Residuals vs. order of data collection")
```



There structure of the data is mostly fine because there is no pattern in the graph other than some high variability at x = 20. Other than this slight concern the structure is adequate.

The residuals are very concerning because many violations are broken. After analyzing the histogram of residuals, the residuals do not appear to be normal. Also the QQ-plot has extremely heavy tails meaning many values appear outside of what is expected. Lastly, after running the Shapiro-Wilk test, I got a P-value of 0.000008 meaning at $\alpha = 0.01667$ I have statistical evidence to suggest the residuals do not form a normal distribution. This means the P-values gained from the linear model may be incorrect. To properly check the P-values, a permutation test is required.

```
reps <- 10
perm_f <- NA
Na_detect <- FALSE

# Run permutation test (checking pour)
for (i in 1:reps) {
  perm_data <- soda_df
  perm_data$score <- sample(perm_data$score)
  perm_model_full <- lm(score ~ pour + consumption + pour * consumption + participant, data = perm_data
  perm_model_reduced <- lm(score ~ consumption + participant, data = perm_data)
  perm_f[i] <- anova(perm_model_reduced, perm_model_full)$F[2]
  if(is.na(perm_f[i])){
    perm_f[i] <- 0
    Na_detect <- TRUE
  }
}
```

```r
obs_f <- anova(soda_no_pour, soda_model)$F[2]

perm_p_value <- sum(perm_f >= obs_f) / reps

perm_test_summary <- data.frame(
  Metric = c("Permutation Pour p-value", "NA Detected"),
  Value = c(perm_p_value, ifelse(Na_detect, "Yes", "No"))
)

kable(perm_test_summary, digits = 4, caption = "Permutation Test Summary for 'pour'")
```

Table 7: Permutation Test Summary for 'pour'

Metric	Value
Permutation Pour p-value	0
NA Detected	No

After running the permutation test to check the pour factor, I still get a P-value of approximately zero, so my original test is correct and I can conclude that the pour variable does affect score. The "NA Detected" Metric is used to test if any of the permutations are the exact same as the original.

```r
reps <- 10
perm_f <- NA
Na_detect <- FALSE

# Run permutation test (checking consumption)
for (i in 1:reps) {
  perm_data <- soda_df
  perm_data$score <- sample(perm_data$score)
  perm_model_full <- lm(score ~ pour + consumption + pour * consumption + participant, data = perm_data
  perm_model_reduced <- lm(score ~ pour + participant, data = perm_data)
  perm_f[i] <- anova(perm_model_reduced, perm_model_full)$F[2]
  if(is.na(perm_f[i])){
    perm_f[i] <- 0
    Na_detect <- TRUE
  }
}

obs_f <- anova(soda_no_consumption, soda_model)$F[2]

perm_p_value <- sum(perm_f >= obs_f) / reps

perm_test_summary <- data.frame(
  Metric = c("Permutation Consumption p-value", "NA Detected"),
  Value = c(perm_p_value, ifelse(Na_detect, "Yes", "No"))
)

kable(perm_test_summary, digits = 4, caption = "Permutation Test Summary for 'consumption'")
```

Table 8: Permutation Test Summary for 'consumption'

Metric	Value
Permutation Consumption p-value	0
NA Detected	No

After running the permutation test to check the consumption factor, I still get a P-value of approximately zero, so my original test is correct and I can conclude that the consumption variable does affect score.

```
reps <- 10
perm_f <- NA
Na_detect <- FALSE

# Run permutation test (checking interaction)
for (i in 1:reps) {
  perm_data <- soda_df
  perm_data$score <- sample(perm_data$score)
  perm_model_full <- lm(score ~ pour + consumption + pour * consumption + participant, data = perm_data
  perm_model_reduced <- lm(score ~ pour + consumption + participant, data = perm_data)
  perm_f[i] <- anova(perm_model_reduced, perm_model_full)$F[2]
  if(is.na(perm_f[i])){
    perm_f[i] <- 0
    Na_detect <- TRUE
  }
}

obs_f <- anova(soda_test_int, soda_model)$F[2]

perm_p_value <- sum(perm_f >= obs_f) / reps

perm_test_summary <- data.frame(
  Metric = c("Permutation Interaction p-value", "NA Detected"),
  Value = c(perm_p_value, ifelse(Na_detect, "Yes", "No"))
)

kable(perm_test_summary, digits = 4, caption = "Permutation Test Summary for 'interaction'")
```

Table 9: Permutation Test Summary for 'interaction'

Metric	Value
Permutation Interaction p-value	0
NA Detected	No

After running the permutation test to check the interaction, I still get a P-value of approximately zero, so my original test is correct and I can conclude that the interaction does affect score.

Discussion

I like discussing

References

Spence, Charles, and Qian Wang. "Sensory expectations elicited by the sounds of opening the packaging and pouring a beverage." Flavour 4 (2015): 1-11

Gonzalez Viejo, Claudia, Damir D. Torrico, Frank R. Dunshea, and Sigfredo Fuentes. "Bubbles, foam formation, stability and consumer perception of carbonated drinks: A review of current, new and emerging technologies for rapid assessment and control." Foods 8, no. 12 (2019): 596.

Wise, Paul M., Madeline Wolf, Stephen R. Thom, and Bruce Bryant. "The influence of bubbles on the perception carbonation bite." PLoS One 8, no. 8 (2013): e71488.